

Practical : 1

AIM:

Linux System Administration and File Management for Marketing Analytics Team

PROBLEM DEFINITION:

Scenario: You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers Where project data, reports, scripts, and user accounts must be managed efficiently. Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team.

Tasks:

1. Identify the current user.
2. Find the current working directory.
3. Display system information.
4. Check system uptime and date.
5. Create a directory named MarketingAnalytics.
6. Create subdirectories: RawData, ProcessedData, Reports.
7. Create data and report files.
8. Add sample content.
9. Display file contents.
10. List all files and directories.
11. Search for report files.
12. Search specific content inside files.
13. Copy report files to Backup directory.
14. Rename files.
15. Move files between directories.
16. Delete a file.
17. Delete an empty directory.

OUTPUT :**Task 1: Identify the Current User**

```
(prince@vbox)-[~]  
$ whoami  
prince
```

Task 2: Find the Current Working Directory

```
(prince@vbox)-[~]  
$ pwd  
/home/prince
```

Task 3: Display System Information

```
(prince@vbox)-[~]  
$ uname -a  
linux vbox 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux
```

Task 4: Check System Uptime and Date

```
(prince@vbox)-[~]  
$ uptime  
18:17:10 up 21 min, 1 user, load average: 0.19, 0.18, 0.17
```

Task 5: Create a Directory Named MarketingAnalytics

```
(prince@vbox)-[~]  
$ mkdir MarketingAnalytics  
  
(prince@vbox)-[~]  
$ ls  
c_prog  Documents  MarketingAnalytics  Pictures  Projects  Templates  
Desktop Downloads  Music              prac3     Public    Videos
```

Task 6: Create Subdirectories

```
(prince@vbox)-[~/MarketingAnalytics]  
$ mkdir RawData ProcessesdData Reports  
  
(prince@vbox)-[~/MarketingAnalytics]  
$ ls  
ProcessesdData RawData Reports
```

Task 7: Create Data and Report Files

```
(prince@vbox)-[~/MarketingAnalytics]
$ touch RawData/customer_data.txt

(prince@vbox)-[~/MarketingAnalytics]
$ touch ProcessedData/cleaned_data.txt
touch: cannot touch 'ProcessedData/cleaned_data.txt': No such file or directory

(prince@vbox)-[~/MarketingAnalytics]
$ touch ProcessesdData/cleaned_data.txt

(prince@vbox)-[~/MarketingAnalytics]
$ touch Report/sales_report.txt
touch: cannot touch 'Report/sales_report.txt': No such file or directory

(prince@vbox)-[~/MarketingAnalytics]
$ touch Reports/sales_report.txt

(prince@vbox)-[~/MarketingAnalytics]
$ find .
.
./ProcessesdData
./ProcessesdData/cleaned_data.txt
./Reports
./Reports/sales_report.txt
./RawData
./RawData/customer_data.txt
```

Task 8: Add Sample Content

```
(prince@vbox)-[~/MarketingAnalytics]
$ echo "CustomerID,Name,Sales" > RawData/customer_data.txt
echo "101,Rahul,25000" >> RawData/customer_data.txt
echo "102,Priya,35000" >> RawData/customer_data.txt

(prince@vbox)-[~/MarketingAnalytics]
$ echo "Cleaned Customer Data" > ProcessesdData/cleaned_data.txt

(prince@vbox)-[~/MarketingAnalytics]
$ echo "Monthly Sales Report" > Reports/sales_report.txt
echo "Total Sales: 60000" >> Reports/sales_report.txt
```

Task 9: Display File Contents

```
(prince@vbox)-[~/MarketingAnalytics]
$ cat RawData/customer_data.txt
CustomerID,Name,Sales
101,Rahul,25000
102,Priya,35000

(prince@vbox)-[~/MarketingAnalytics]
$ cat ProcessedData/cleaned_data.txt
cat: ProcessedData/cleaned_data.txt: No such file or directory

(prince@vbox)-[~/MarketingAnalytics]
$ cat ProcessesdData/cleaned_data.txt
Cleaned Customer Data

(prince@vbox)-[~/MarketingAnalytics]
$ cat Reports/sales_report.txt
Monthly Sales Report
Total Sales: 60000
```

Task 10: List All Files and Directories

```

(prince@vbox)-[~/MarketingAnalytics]
$ ls
ProcessedData  RawData  Reports

(prince@vbox)-[~/MarketingAnalytics]
$ ls -l
total 12
drwxrwxr-x 2 prince prince 4096 Aug  4 18:22 ProcessedData
drwxrwxr-x 2 prince prince 4096 Aug  4 18:21 RawData
drwxrwxr-x 2 prince prince 4096 Aug  4 18:23 Reports

(prince@vbox)-[~/MarketingAnalytics]
$ ls -R
.:
.: Integer Multiplication
ProcessedData  RawData  Reports

./ProcessedData:
cleaned_data.txt

./RawData:
customer_data.txt

./Reports:
sales_report.txt

```

Task 11: Search for Report Files

```

(prince@vbox)-[~/MarketingAnalytics]
$ find . -name "*report*"
./Reports/sales_report.txt

```

Task 12: Search Specific Content Inside Files

```

(prince@vbox)-[~/MarketingAnalytics]
$ grep "Sales" Reports/sales_report.txt
Monthly Sales Report
Total Sales: 60000

(prince@vbox)-[~/MarketingAnalytics]
$ grep "Rahul" RawData/customer_data.txt

```

Task 13: Copy Report Files to Backup Directory

```

(prince@vbox)-[~/MarketingAnalytics]
$ mkdir Backup

(prince@vbox)-[~/MarketingAnalytics]
$ cp Reports/sales_report.txt Backup/

(prince@vbox)-[~/MarketingAnalytics]
$ ls Backup
sales_report.txt

```

Task 14: Rename Files

```
(prince@vbox)-[~/MarketingAnalytics]
$ mv Reports/sales_report.txt Reports/final_report.txt

(prince@vbox)-[~/MarketingAnalytics]
$ ls Reports
final_report.txt
```

Task 15: Move Files Between Directories

```
(prince@vbox)-[~/MarketingAnalytics]
$ mv ProcessesdData/cleaned_data.txt Reports/

(prince@vbox)-[~/MarketingAnalytics]
$ ls Reports
cleaned_data.txt  final_report.txt
```

Task 16: Delete a File

```
(prince@vbox)-[~/MarketingAnalytics]
$ rm RawData/customer_data.txt

(prince@vbox)-[~/MarketingAnalytics]
$ ls RawData
```

Task 17: Delete an Empty Directory

```
(prince@vbox)-[~/MarketingAnalytics]
$ rmdir RawData

(prince@vbox)-[~/MarketingAnalytics]
$ ls
Backup  ProcessesdData  Reports
```

KEY QUESTIONS :**Q1. What is the difference between absolute and relative paths?****Answer:**

An absolute path gives the complete location of a file or directory starting from the root (/) directory. It always points to the same location regardless of the current directory. A relative path gives the location of a file based on the current working directory. It is shorter and is commonly used when working inside the same project.

Q2. Why are Linux permissions important for file security?**Answer:**

Linux permissions help protect files and directories from unauthorized access. They control who can read, write, or execute a file, ensuring that only authorized users can modify important data. This improves system security and prevents accidental or malicious changes.

Q3. What is the purpose of the grep command?**Answer:**

The grep command is used to search for specific words or patterns inside files. It quickly finds matching lines in one or more files, making it easier to locate information in large text files, logs, or reports.

Q4. How does Linux support efficient file management?**Answer:**

Linux provides many built-in commands to create, copy, move, rename, search, and delete files and directories. It also offers a hierarchical directory structure and powerful command-line tools, making file management fast, organized, and efficient.

Q5. Why are backups important in enterprise environments?**Answer:**

Backups are important because they protect valuable data from accidental deletion, hardware failure, cyberattacks, or system crashes. They allow organizations to restore lost data quickly, reduce downtime, and ensure business operations continue without major interruptions.

CONCLUSION :

This practice helped us understand the basic Linux commands used for system administration and file management. We learned how to create and organize directories, manage files, search for data, and create backups. These skills are essential for maintaining secure, organized, and efficient Linux systems in real-world enterprise environments.

GITHUB : https://github.com/Devsavani12/BIG-DATA_SHP